

strategies adopted by attackers.

Kali Linux: Most tried and tested approaches to penetration testing use the legendary Kali Linux system. Kali offers an entire operating system dedicated to the single task of emulating an infiltrator within the network. It is



Figure 3: Kali Linux (Credits: kali.org)

supported by a robust development environment with thousands of packages available for use. Recent releases have focused on addressing the much-hyped Spectre and Meltdown vulnerabilities and now support hardware-level encryption of sorts.

Bro IDS: Marketed as the software that has successfully brought together research and implementation, Bro is an intrusion detection system that has been developed at Berkeley and the National Centre for Supercomputing Applications at Urbana-Champaign, Illinois, USA. Focusing on network security and monitoring, it provides a comprehensive platform for network traffic analysis.

Wireshark: Running on Windows, OS X, and UNIX, Wireshark is one



Figure 4: Wireshark (Credits: howtouselinux.net)

of the most effective network packet analysers. It offers a variety of capturing and analysing options, along with a well-designed interface to visualise and view the contents and details of the captured data. Alternatives that focus on the command line execution include Tshark and Tcpdump.

Social Engineering Toolkit (SET): One of the more fun tools to play around with, the SET presents, quite literally, a menu of options that can be leveraged to exploit the human

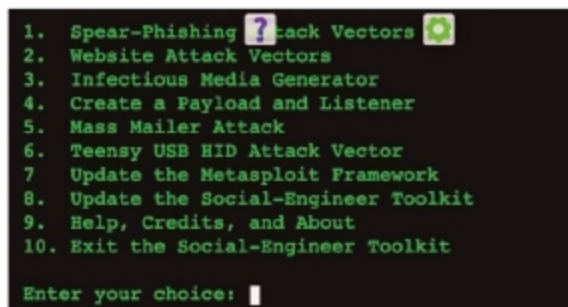


Figure 5: Social Engineering Toolkit

factors that impact the system. It focuses on phishing, malware delivered via infected PDF files and other social engineering practices that can be used to bypass the security rules within a system. Using SET, security teams can design and deploy internal attacks on members of the organisation in order to test the security awareness and resilience of the network.

The new entrants

We now shift our focus to a few noteworthy tools that caught our attention based on their popularity in the open source community.

WiPi Hunter: This is designed to detect illegal wireless network activities performed by special software and hardware, and comprises various modules involving fake access points for monitoring and analysis, detection of attacks using KARMA, apart from calculating a wireless security score.

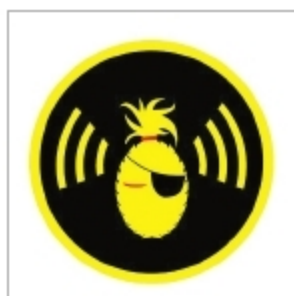


Figure 6: WiPi Hunter (Credits: toolswatch.org)

WHID or WiFi-HID Injector: Since the initial appearance of HID attacks, many tools and devices to guard against them have been released. However, offensive security researchers were always



Figure 7: WHID Injector (Credits: toolswatch.org)

seeking cheap and dedicated hardware that could be controlled remotely and that's how the WHID Injector was born. It has been designed using an Atmega 32u4 and an ESP-12, which means it can be easily controlled over the Wi-Fi network and potentially bypass air-gapped environments.

Leviathan Framework: Leviathan is a mass-audit toolkit comprising



Figure 8: Leviathan Framework (Credits: toolswatch.org)

various features including SQL injection, custom exploits, brute force attacks and wide range service discovery. It offers the flexibility of using tools like dss, masscan and ncrack, or a combination of these. The goal is to audit as many systems as possible within a wide IP range.

LuLu: As its website states, the world has turned into a Web of



Figure 9: LuLu (Credits: hehacktoday.com)

connected devices. And often, the root cause of these problems is the very connection that initiated the Web. LuLu presents a fresh perspective; it is a firewall that aims to block all outgoing connections unless explicitly approved by the user. **END** 🐧

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